

Shopify Sales Analysis

Business Intelligence Project Brief • An Interactive Power BI Dashboard

Tool	Power BI Desktop	Skills	Power Query (ETL), Data Modeling, DAX
Focus	Sales, customers, retention	Dataset	Shopify sales / transactions export

In this project you will analyse Shopify sales data in **Power BI** to uncover meaningful insights into transaction performance, customer purchasing behaviour, and long-term customer value. Your goal is to design an **interactive dashboard** that helps stakeholders spot patterns in revenue generation, customer retention, and engagement — so they can make data-driven decisions. Work through the stages below and answer every question with visuals, DAX measures, and short written explanations.

***What to submit:** (1) your Power BI file (.pbix), (2) a short written report answering the questions below, and (3) a one-page exported view or screenshots of your final dashboard. For every measure you create, include the DAX and a one-line explanation of what it calculates.*

STAGE 0 Problem Statement

1. In your own words, state the **goal** of this project and who the **stakeholders** are. What decisions should the dashboard help them make?
2. Write at least **five business questions** your dashboard will answer — covering revenue, purchasing behaviour, and customer retention.
3. List the **key metrics (KPIs)** you expect to track (for example: total revenue, number of orders, average order value, repeat-purchase rate, customer lifetime value).
4. State the main **objective** — the single deliverable you are producing — and three steps to reach it.

STAGE 1 Data Preparation & Modeling

1.1 Import & Clean (Power Query)

1. Import the Shopify data into Power BI. How many tables and rows do you have? Briefly describe the key columns (order ID, customer ID, order date, product, quantity, price, etc.).
2. Use Power Query to clean the data: handle missing values, fix data types (especially dates and numbers), remove duplicates, and standardise text. Document each step you applied and why.
3. Create any helper columns you will need — for example order month/year, or a flag for first vs. repeat orders.

1.2 Data Model

1. Design your **data model**. Will you use a star schema with fact and dimension tables? Draw or describe the relationships between your tables.
2. Add a dedicated **Date (calendar) table** and mark it as a date table. Why is this important for time-based analysis?

STAGE 2 Transaction Performance Analysis

1. Create DAX measures for **Total Revenue**, **Total Orders**, and **Average Order Value (AOV)**. Display them as KPI cards.
2. Visualise **revenue over time** (by month/quarter/year). Are there clear trends, seasonal peaks, or dips?
3. Break revenue down by **product or category**. Which products drive the most revenue, and which underperform?
4. Analyse orders by **time of day or day of week** (if available). When do customers buy most? What could a store do with this?
5. Add a measure showing revenue growth versus the previous period (e.g. month-over-month). How is the business trending?

STAGE 3 Customer Purchasing Behaviour

1. How many **unique customers** are there? What share of customers are **new** versus **returning**?
2. Analyse **order frequency**: how many orders does a typical customer place? Visualise the distribution.
3. Identify your **top customers** by total spend. What share of total revenue do the top 10% (or top 20%) generate?
4. Segment customers by behaviour — for example using an RFM approach (Recency, Frequency, Monetary) or simple spend tiers. Describe the segments you create.
5. What products are most popular among returning customers versus first-time buyers?

STAGE 4 Retention & Long-Term Customer Value

1. Define **customer retention** for this business. Create a measure for the **repeat-purchase rate** (the share of customers who order more than once).
2. Build a simple **cohort analysis**: group customers by the month of their first purchase and track how many return in later months. What does the retention pattern look like?
3. Estimate **Customer Lifetime Value (CLV)** — for example average order value × purchase frequency × customer lifespan. Which customer segment has the highest CLV?
4. How long, on average, is the gap between a customer's first and second purchase? What does this suggest about when to re-engage them?
5. Based on retention and CLV, which customers should the business prioritise keeping, and how?

STAGE 5 Dashboard Design & Insights

1. Assemble your visuals into a clean, interactive dashboard. Add **slicers/filters** (e.g. by date range, product category, customer segment) so users can explore the data themselves.
2. Organise the layout so a stakeholder can read the story top-to-bottom: headline KPIs first, then trends, then detail. Explain your layout choices.
3. Ensure the report is interactive — test that clicking one visual cross-filters the others. Confirm this works.
4. Write **three to five key insights** a stakeholder should take away from your dashboard, each stated in plain language and backed by a specific visual.
5. Turn your insights into **business recommendations**: what concrete actions should the company take to grow revenue and improve retention?

Presentation tip: Imagine presenting to a store owner who does not use Power BI. Every chart should answer a business question at a glance, and every recommendation should follow directly from what the data shows — not from assumptions.